Absolute forecasting error by routine-pattern deviation

H6 (30 min) · full test set · training-fitted scenario thresholds

(a) Chengdu · inflow

	Low	Moderate	High	Extreme
HA	1.08	0.93	0.88	1.52
SVR	0.33	0.50	0.80	1.46
VAR	0.40	0.57	0.85	1.46
DCRNN	0.41	0.42	0.74	1.41
STGCN	0.16	0.35	0.71	1.38
GWNet	0.16	0.35	0.72	1.39
STNorm	0.15	0.34	0.70	1.35
DSTAGNN	0.89	0.83	0.89	1.41
PDFormer	0.75	0.96	1.47	3.23
STWave	0.15	0.34	0.69	1.35
PatchTST	0.54	0.39	0.76	1.49
TimeMixer	0.56	0.42	0.79	1.49
STDN	1.11	1.16	1.05	1.42
MetaDG	1.05	1.03	1.05	1.53
HyperD	0.16	0.35	0.71	1.39
RAST	0.16	0.35	0.72	1.40
CMSRN	0.16	0.34	0.71	1.34

(b) Chengdu · outflow

	Low	Moderate	High	Extreme
HA	1.05	0.91	0.90	1.37
SVR	0.30	0.51	0.84	1.32
VAR	0.37	0.57	0.87	1.33
DCRNN	0.35	0.43	0.78	1.27
STGCN	0.14	0.38	0.77	1.27
GWNet	0.14	0.38	0.77	1.27
STNorm	0.13	0.37	0.75	1.24
DSTAGNN	0.91	0.86	0.94	1.31
PDFormer	0.70	0.98	1.54	2.75
STWave	0.13	0.37	0.76	1.24
PatchTST	0.50	0.42	0.81	1.35
TimeMixer	0.52	0.44	0.83	1.36
STDN	1.10	1.13	1.03	1.30
MetaDG	1.04	1.03	1.07	1.39
HyperD	0.14	0.38	0.76	1.26
RAST	0.14	0.38	0.78	1.28
CMSRN	0.14	0.38	0.76	1.25

(c) Shanghai · inflow

	Low	Moderate	High	Extreme
HA	1.02	0.88	0.97	1.49
SVR	0.36	0.65	0.93	1.40
VAR	0.45	0.72	0.95	1.41
DCRNN	0.20	0.55	0.87	1.34
STGCN	0.18	0.54	0.86	1.35
GWNet	0.18	0.54	0.86	1.34
STNorm	0.18	0.53	0.84	1.30
DSTAGNN	0.83	0.80	0.93	1.37
PDFormer	0.69	1.13	1.66	3.07
STWave	0.18	0.53	0.84	1.30
PatchTST	0.42	0.57	0.90	1.44
TimeMixer	0.43	0.60	0.93	1.44
STDN	1.15	1.16	1.14	1.53
MetaDG	1.05	1.07	1.13	1.53
HyperD	0.19	0.54	0.85	1.33
RAST	0.18	0.54	0.86	1.34
CMSRN	0.18	0.54	0.85	1.30

(d) Shanghai · outflow

	Low	Moderate	High	Extreme
HA	1.01	0.89	0.99	1.68
SVR	0.34	0.66	0.95	1.59
VAR	0.45	0.72	0.97	1.60
DCRNN	0.18	0.56	0.88	1.53
STGCN	0.16	0.56	0.88	1.53
GWNet	0.16	0.56	0.88	1.53
STNorm	0.16	0.55	0.86	1.49
DSTAGNN	0.84	0.81	0.96	1.53
PDFormer	0.68	1.16	1.74	3.59
STWave	0.16	0.55	0.86	1.48
PatchTST	0.41	0.59	0.92	1.64
TimeMixer	0.42	0.61	0.95	1.64
STDN	1.10	0.97	1.01	1.58
MetaDG	1.03	1.10	1.21	1.77
HyperD	0.16	0.55	0.87	1.50
RAST	0.16	0.56	0.88	1.54
CMSRN	0.16	0.55	0.87	1.48



MAE (original flow units; lower is better)

Shared absolute-error color scale across all panels. CMSRN uses the existing Full checkpoint.

All finite values are shown without clipping; gray cells indicate an empty or undefined group.